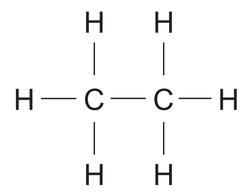
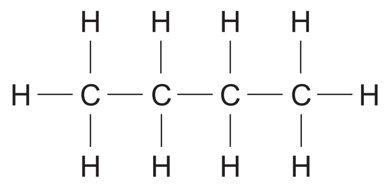
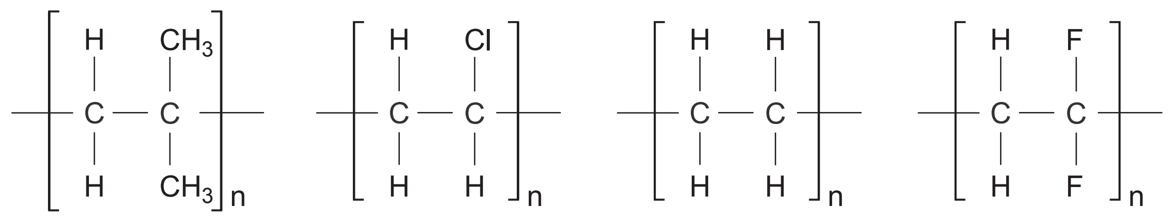
3. (a) Complete the table. [2]

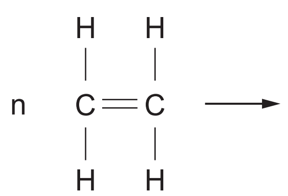
Name	Molecular formula	Structural formula
ethane	C_2H_6	
propane	C_3H_8	
butane	....	

(b) The box contains four polymers **A**, **B**, **C** and **D**.



A **B** **C** **D**

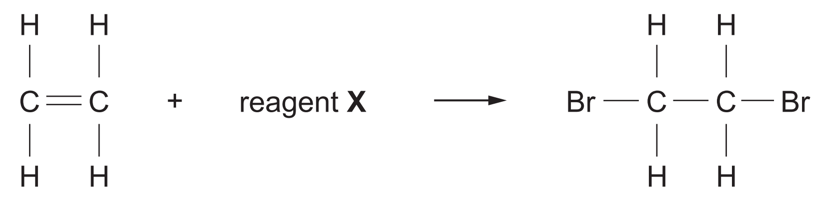
Give the **letter** of the polymer formed during the polymerisation of ethene. [1]



Letter ....

Examiner only

(c) The equation shows the reaction taking place when testing for a C═C bond.



colourless orange colourless

bromine water	**potassium dichromate**	**universal indicator**	**barium chloride**

Choose the name of reagent **X** from the box above.

.... [1]

(d) The equation shows the fermentation of glucose.

$$C_6H_{12}O_6 \xrightarrow{\text{yeast}} 2C_2H_5OH + 2CO_2$$

(i) Underline the chemical name for C_2H_5OH. [1]

sugar **biofuel** **ethanol** **alcohol**

(ii) Calculate the relative molecular mass, M_r, of C_2H_5OH. [2]

$A_r(H) = 1$ $A_r(O) = 16$ $A_r(C) = 12$

M_r = ...

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 (3410U20-1) **Turn over.**